ASEAN
GEOAI FUSION 2026

Module 7:
Advanced Knowledge: Advanced Geospatial Techniques in GeoAI Applications (AD1002)

AGAIF 2026 Boot Camp 2
Level: Advanced Knowledge of GeoAI

Thursday | 23 July 2026

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Universiti Tenaga Nasional (UNITEN)



Session 6:

Bootcamp 2 Synthesis

Evidence-to-Data Workflow

Show how research-supported factors became variables and datasets for the Sarawak connectivity assessment workflow.

Canvas Reference:
C5 GeoAI Intelligence

1 1. Evidence Base



- Scopus AI problem discovery
- verified research findings
- ASEAN Digital Masterplan 2030
- ITU and policy evidence

Research and policy evidence identify the factors that matter.

2 2. Analytical Factors

- performance quality
- observational strength
- settlement presence
- population context
- public-service context

These become the logic for screening and assessment.

3 3. Variables

Performance:
- avg_d_mbps
- avg_u_mbps
- avg_lat_ms

Evidence strength:
- tests
- devices

Context:
- name / place
- est_population_tile
- service_group

Each factor is converted into measurable analysis fields.

4 4. Datasets & Layers



- Ookla mobile performance tiles (parquet)
- OSM mapped settlements
- WorldPop 100 m population raster
- OSM public-service facilities

Variables are linked to real datasets and usable spatial layers.

Output of this step

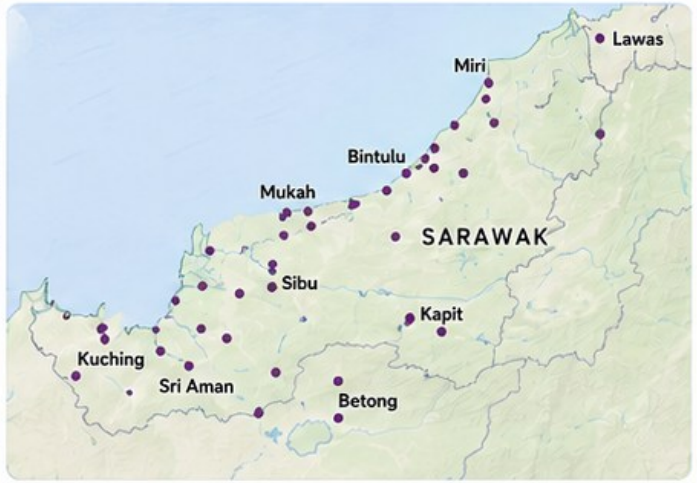
An analysis-ready data blueprint that links evidence, factors, variables, and datasets for Canvas 5 design work.

GeoLibre Exploration

Load OSM mapped settlements and WorldPop layers to compare mapped communities with population evidence in Sarawak.

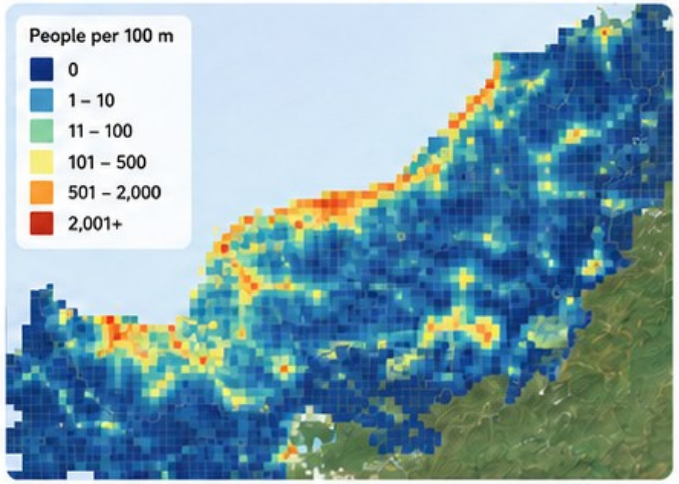
Canvas Reference:
C5 GeoAI Intelligence

1 Layer 1 — OSM mapped settlements



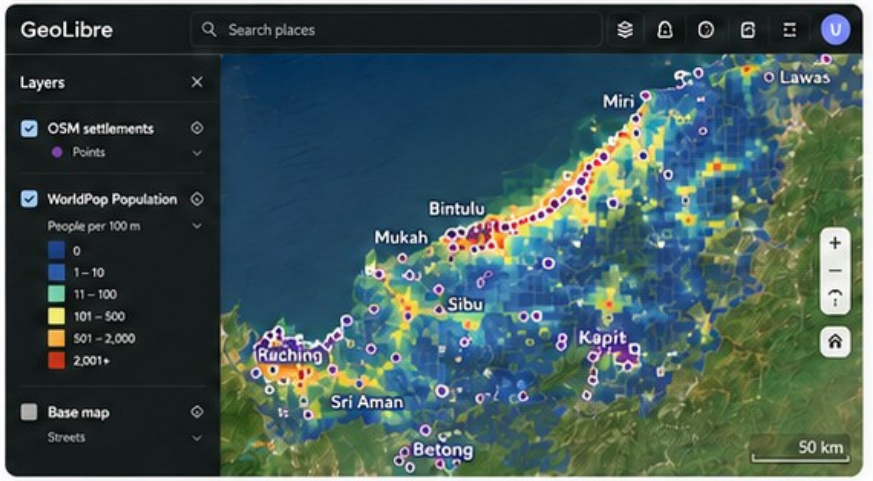
- Represents mapped towns, villages and hamlets from OpenStreetMap
- Useful for locating named communities
- Helps define the decision unit for later matching

2 Layer 2 — WorldPop population grid



- Provides gridded population estimates
- Shows where more people may be concentrated
- Adds context beyond mapped place names alone

3 What users observe in GeoLibre



- Communities and population are related, but not identical
- Some mapped settlements sit in lower-population areas
- Some higher-population cells include multiple nearby communities
- This comparison prepares the evidence for later screening

Key interpretation

GeoLibre helps participants visually compare where communities are mapped and where population is estimated. At this stage, the goal is evidence exploration and spatial understanding — not yet performance screening.

Add Ookla Mobile Performance

Understand what the Ookla data represents and why no observation means no coverage—not bad performance.

Canvas Reference:
C5 GeoAI Intelligence

1 What is Ookla Open Data?

OOKLA OPEN DATA

- Public dataset from Speedtest measurements
- Collected from users on mobile and fixed broadband
- Includes real-world performance metrics with location and time

Key value
Provides ground-truth performance evidence across the area based on real user experience.

2 Key Ookla Fields We Use

Field	Description
download (Mbps)	Download speed
upload (Mbps)	Upload speed
latency (ms)	Network latency
tests	Number of speed tests in the tile
devices	Number of unique devices in the tile
timestamp	When the test was performed
latitude / longitude	Location of the test

3 No Observation ≠ No Coverage

Important:
No test in a tile does NOT mean the area has poor performance.

It usually means:

- No users tested in that area yet
- Low population or low network usage
- Remote or hard-to-reach location

Key takeaway:
We use Ookla to understand measured performance —not the absence of data.

Key interpretation

Ookla data shows where real users experience the network. It is a critical layer for identifying areas with limited or lower performance — and a foundation for evidence-based screening.

Filter Global Ookla to Show Sarawak

Use Colab to filter the global dataset and focus only on Sarawak tiles for our analysis.

Canvas Reference:
C5 GeoAI Intelligence

1 Global Ookla Dataset

3,186,269
global records
from Ookla Open Data

- Contains speed tests from around the world
- Covers many countries and regions
- Needs to be filtered to our study area

Goal: Keep only the records that are located in Sarawak.

2 Colab Filtering Process

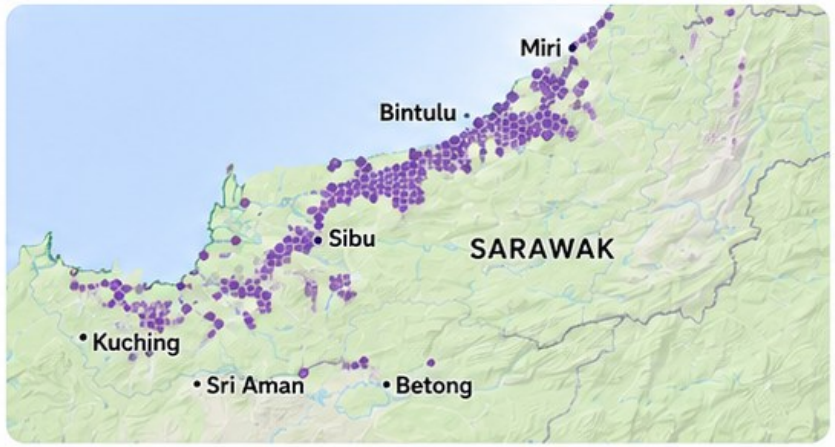
Load the data
Import the global Ookla dataset into Google Colab.

Filter by location
Keep only records where latitude and longitude fall within Sarawak boundary.

Result
Filtered dataset ready for further analysis in Session 6.

Output:
5,912 records (Sarawak tiles)

3 Where Are These Records?



These 5,912 records represent real user experiences within Sarawak and will be the foundation for performance analysis and screening.

Key takeaway

We reduce the global dataset to Sarawak-only records using Colab.
This ensures our analysis is focused, relevant and decision-ready.

Classify Download Performance

Quantile classification to identify the lowest-performing areas

We classify download speed using quantiles to focus on the lowest 20% of performance.

Canvas Reference:
C5 GeoAI Intelligence

1 Quantile Classification

We use quantiles to divide download speed into five equal groups.

Q1 (Lowest 20%)	Lowest performance
Q2 (20–40%)	Below average
Q3 (40–60%)	Average
Q4 (60–80%)	Above average
Q5 (Highest 20%)	Highest performance

Focus: Q1 (Lowest 20%)
These are the areas with the slowest download speeds.

2 Threshold for Lowest 20%

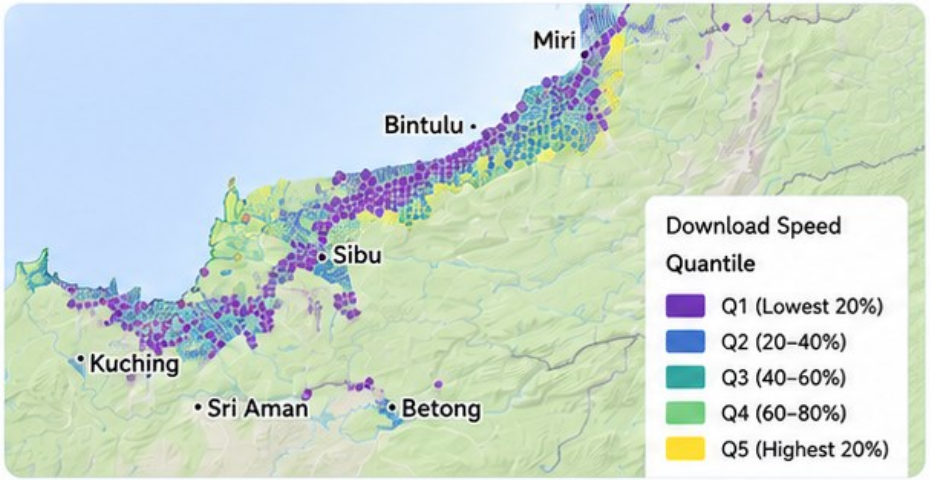
23.53 Mbps
Download speed threshold for the lowest 20% (Q1)

Interpretation:
Tiles with download speed ≤ 23.53 Mbps are classified as lowest-performing areas.

Why quantiles?

- Not affected by extreme values
- Ensures equal number of tiles per class
- Ideal for data-driven, relative comparison

3 Lowest 20% Download Performance (Q1)



Purple areas are the lowest-performing tiles.
These are our priority areas for further screening.

Key takeaway

Using quantile classification, we identify the lowest 20% of download performance.
Tiles with download speed ≤ 23.53 Mbps are flagged as lowest-performing areas for deeper analysis.

Evidence Strength Screening

Use tests and devices to keep only tiles with stronger evidence

We apply a baseline rule to ensure that selected tiles have enough user activity to be reliable for further analysis.

Canvas Reference:
C5 GeoAI Intelligence

1 Evidence Strength Rule

We keep only tiles that meet the minimum evidence threshold for user activity.

Baseline rule:
≥15 tests and ≥5 devices

- Ensures data is based on sufficient user activity
- Reduces noise from tiles with very few measurements
- Improves reliability of performance comparison

Goal: Focus on areas where real users have generated enough evidence to reflect actual network performance.

2 How We Apply the Rule

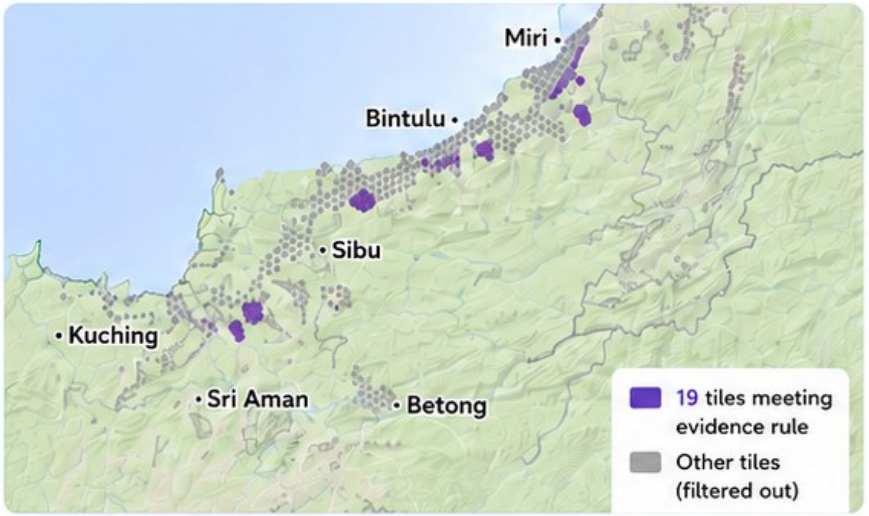
Check each tile
Evaluate the number of tests and devices in each tile.

Apply the threshold
Keep tiles with ≥15 tests and ≥5 devices.

Result
Tiles that meet the rule are kept for the next step.

Outcome:
19 lower-performance tiles with stronger evidence

3 Tiles That Meet the Evidence Rule



These 19 tiles have enough user activity and will be used to identify communities for further analysis.

Key takeaway

Applying the evidence strength rule helps us focus only on tiles with sufficient user activity, resulting in 19 lower-performance tiles that are reliable for the next stage of analysis.

Screening Result

From lowest-performance tiles to stronger-evidence tiles

After applying the evidence strength rule, we reduce the number of tiles and keep those with more reliable user activity.

Canvas Reference:
C5 GeoAI Intelligence

1 Before vs After Evidence Screening

We refine the list by keeping only tiles that meet the evidence strength rule (≥15 tests and ≥5 devices).

Stage	Number of Tiles
Lowest-performance tiles (Q1) Lowest 20% by download speed	1,183
Tiles meeting evidence rule ≥15 tests and ≥5 devices	19
Reduction Fewer tiles, stronger evidence	98.4%

Outcome: We focus on fewer tiles with stronger and more reliable evidence.

2 Evidence Screening Impact

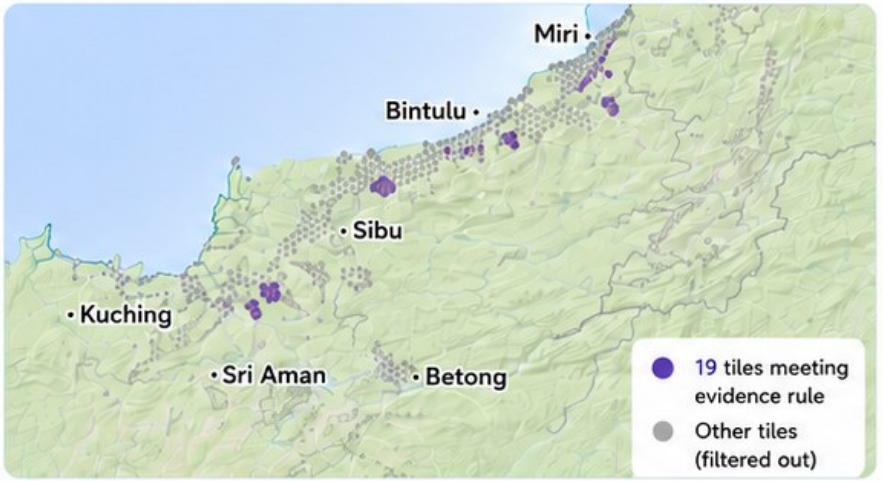
Applying the rule removes tiles with very low user activity.



98.4% reduction
Focus on quality, not just quantity.

3 Where Are the 19 Tiles?

These tiles are distributed across Sarawak.



These 19 tiles will be used for the next step: matching to communities and adding population context.

KEY TAKEAWAY

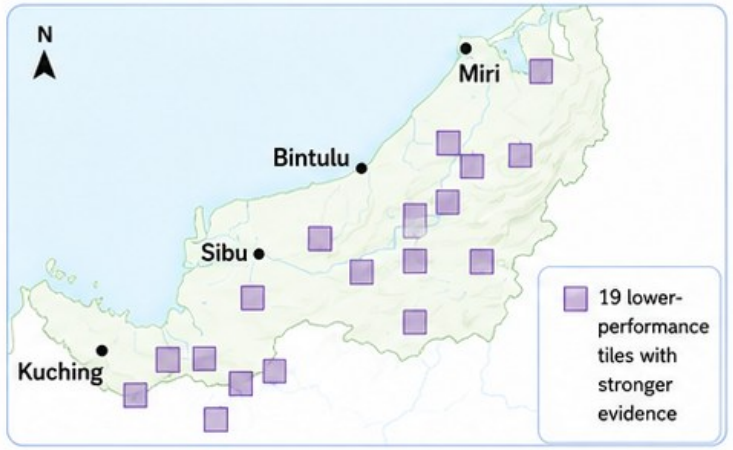
We start with 1,183 lowest-performance tiles but keep only 19 tiles with stronger evidence.
This ensures our analysis is based on tiles with sufficient user activity and more reliable measurements.

Match to Communities

Use spatial matching to connect screened lower-performance tiles with mapped settlements in Sarawak.
This step translates tile-level evidence into named communities for further assessment.

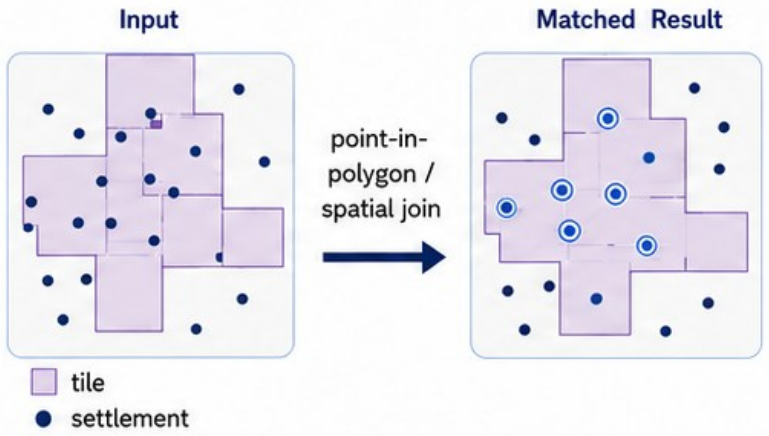
Canvas Reference:
C5 GeoAI Intelligence

1 Selected Screened Tiles



- Input layer from the screening step
- Represents reliable lower-performance tile evidence
- Used as the spatial layer for matching

2 Spatial Join with Mapped Settlements



- Check which mapped settlements fall inside the selected tiles
- Link each matched community to tile attributes
- Convert tile evidence into named places

3 Candidate Communities Identified

Rank	Candidate Community
1	Kampung Hulu
2	Kampung Seberang Maludam
3	Kampung Seberang Sebuyau
4	Kampung Tebelu
5	Maludam
6	Beladin
7	Kampung Simpok

Result: 7 candidate communities

- Named communities can now be reviewed directly
- Supports prioritisation for further assessment
- Prepares the evidence for contextual enrichment

Key takeaway

Spatial matching turns 19 screened lower-performance tiles into 7 named candidate communities, making the analysis easier to interpret, communicate and assess.

Bring in Population Context

Understand how many people may be affected in each low-performance area

We overlay population data to estimate the number of people potentially affected in each of the 19 lower-performance tiles.

Canvas Reference:
C5 GeoAI Intelligence

1 Why Population Matters

People impact
Low performance in highly populated areas affects more people.

Better prioritisation
Helps us prioritise areas where intervention can benefit more communities.

Data source
Population data from WorldPop (2020) at ~100m resolution.

Goal: Estimate the population potentially impacted in each of the 19 lower-performance tiles.

2 What We Do

Overlay population
Intersect the 19 tiles with the population grid.

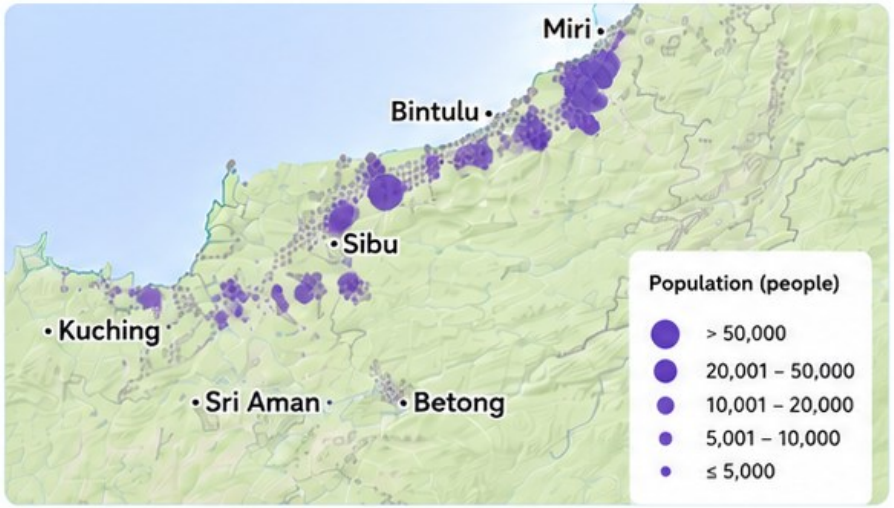
Calculate population
Sum the population within each tile.

Get total estimate
Estimate the total population potentially affected.

Outcome:

~480,387
people potentially affected across the 19 lower-performance tiles.

3 Population Potentially Affected



The size of each circle represents the estimated population in each of the 19 lower-performance tiles.

Key takeaway

Adding population context shows us where more people may be affected.
These insights help us focus on areas where connectivity gaps matter most.

Public-Service Extension

Add schools, clinics, hospitals and emergency facilities from OSM to enrich impact context around candidate communities in Sarawak.

Canvas Reference:
C5 GeoAI Intelligence

1 Why We Add Public-Service Context



- helps interpret potential community importance beyond speed alone
- shows where connectivity may support education, healthcare and emergency response
- adds service-access context for later discussion

Purpose: strengthen impact interpretation, not automated prioritisation.

2 Service Layers Added from OSM



School / education facility



Clinic / health clinic



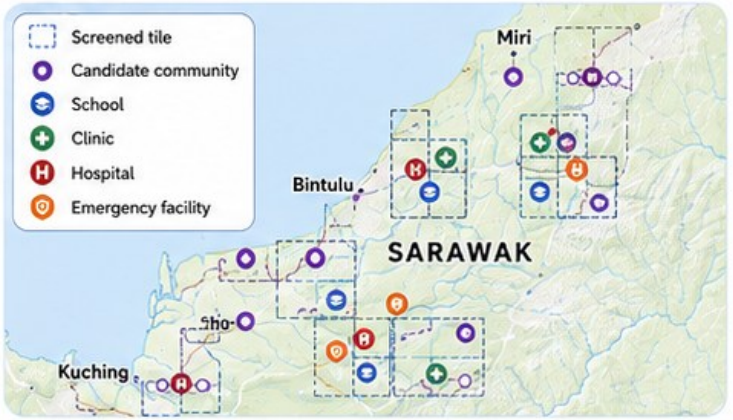
Hospital / major healthcare facility



Emergency / police, fire or ambulance-related facility

- extracted as mapped supporting facilities
- linked spatially to screened areas and candidate communities
- used as contextual evidence only

3 How to Read the Extended Map



- candidate communities remain the main decision focus
- public-service points help explain possible social value
- services do not automatically increase ranking score
- findings support impact storytelling and further assessment

Example use: Identify whether a lower-performing area also serves nearby schools or health facilities.

Key takeaway Public-service layers extend the evidence from network performance toward community impact. They help users see why some candidate areas may matter more for service access, while remaining supporting context rather than a direct ranking weight.

Interactive Prototype

Folium HTML map: 19 screened tiles + 7 ranked communities + popups

We visualise the results in an interactive map to explore locations, evidence and population context in one place.

Canvas Reference: Canvas 6

1 Prototype Highlights

19 screened tiles
Lower-performance tiles with stronger evidence.

7 ranked communities
Matched from OSM settlements with population context.

Interactive popups
View key attributes and population information for each feature.

Multiple layers
Toggle layers to explore performance, communities and context.

Goal: Provide an easy-to-explore map that supports evidence-based decision making.

2 What's on the Map?

Lower-performance tiles (19)
Tiles classified in the lowest 20% download performance.

Screened tiles (stronger evidence)
Tiles that meet the evidence strength rule (≥15 tests and ≥5 devices).

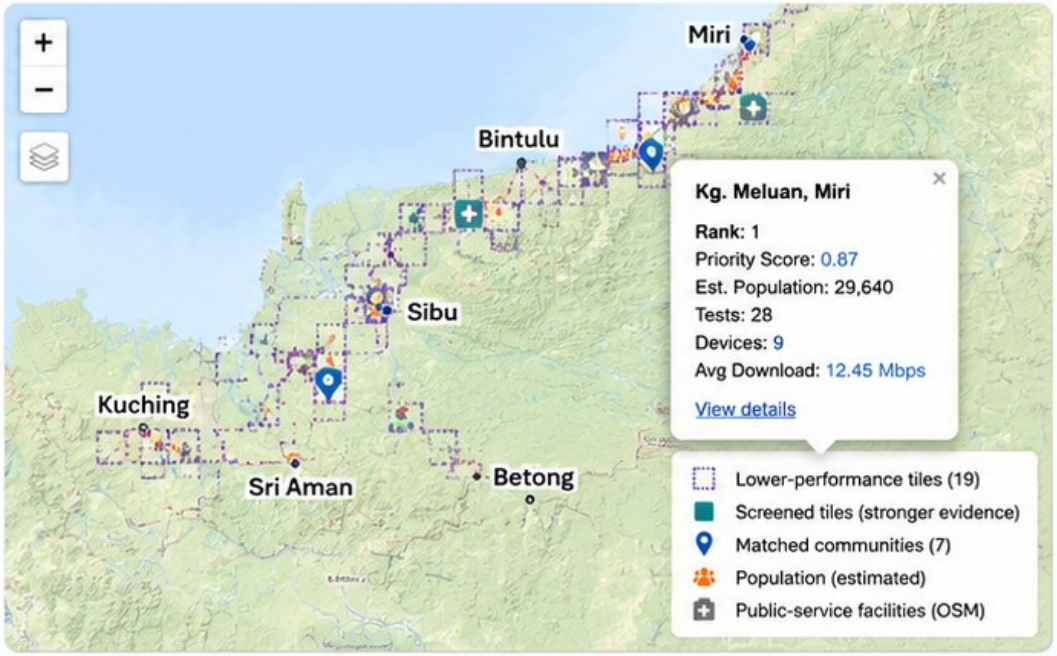
Matched communities (7)
OSM settlements matched to screened tiles and ranked.

Population (estimated)
WorldPop estimated population within each tile.

Public-service facilities (OSM)
Schools, clinics, hospitals and emergency facilities (context only).

Outcome:
An interactive view to explore 19 screened tiles and 7 ranked communities with population context and popups.

3 Prototype Preview (Example)



Use the map to explore locations, click on features to view details, and understand the communities behind the data.

Tools & Stack

Use a connected toolchain to move from evidence gathering to spatial analysis and an interactive prototype.

Canvas Reference:
C7 Technology Stack

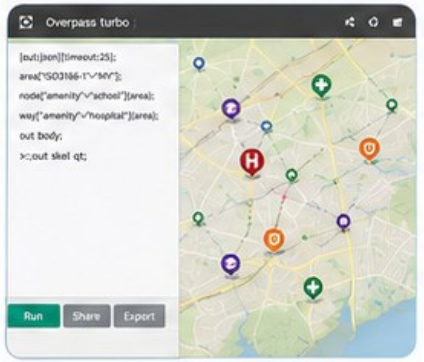
This session combines research, policy evidence, data extraction, mapping, analysis and delivery tools into one reproducible GeoAI workflow.

1 Research & Policy Grounding



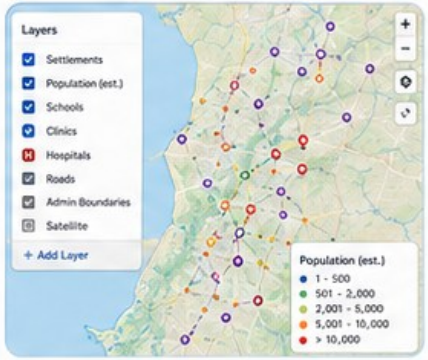
- Scopus AI
- ASEAN / ITU evidence
- identify problem factors and current context

2 Data Extraction



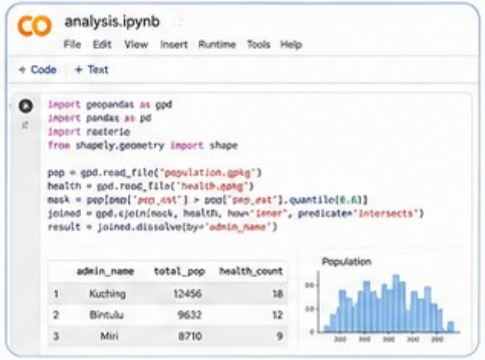
- Overpass Turbo
- OpenStreetMap queries
- mapped settlements and public-service layers

3 Exploration & Layer Review



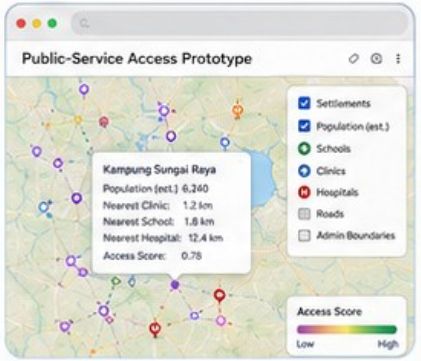
- GeoLibre
- compare settlements, population and performance layers
- inspect spatial patterns before screening

4 Processing & Analysis



- Colab / Python
- GeoPandas and Rasterio
- filtering, quantiles, spatial joins and overlays

5 Prototype Output



- Folium HTML map
- popups and layered outputs
- shareable prototype for demonstration

Key takeaway | reproducible workflow | transparent analytical steps | modular tools that work together

What We Achieved in Session 6

From evidence screening and contextual analysis to a defensible prototype workflow

Session 6 transformed the Session 5 problem definition into a data-driven GeoAI screening workflow and prototype direction.

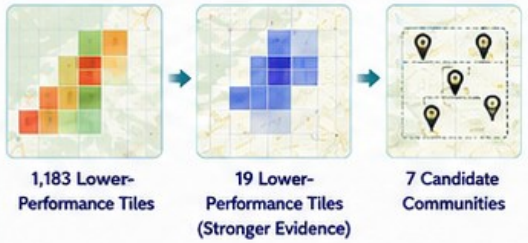
Canvas Progress:
- C5 GeoAI Intelligence
- C6 GeoAI Solution Design
- C7 Technology Stack
- C8 Responsible GeoAI

1 C5 — Evidence-to-Data Workflow



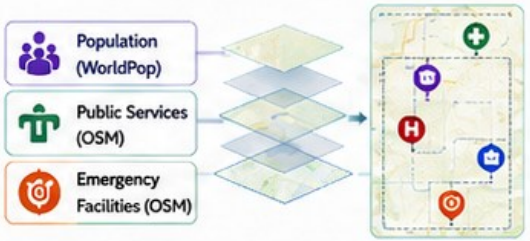
- Converted research-supported factors into variables and datasets.
- Loaded OSM settlements, WorldPop population grid, and Ookla mobile performance data.
- Filtered 3,186,269 global Ookla records to 5,912 Sarawak tiles.
- Classified lower performance using the lowest 20% threshold: 23.53 Mbps.

2 C5 — Screening and Matching Results



- Applied the evidence rule: ≥15 tests and ≥5 devices.
- Narrowed 1,183 lowest-performance tiles to 19 lower-performance tiles with stronger evidence.
- Matched screened tiles with OSM mapped settlements using spatial join.
- Identified 7 candidate communities for further assessment.

3 C5 — Context Layers Added



- Added WorldPop population context to estimate potentially affected population.
- Added OSM public-service context: schools, clinics, hospitals, and emergency facilities.
- Used contextual layers to support interpretation, not automatic ranking.
- Reinforced that the workflow supports prioritisation for further assessment.

4 C6–C8 — Prototype, Stack and Responsible GeoAI



- Produced a Folium HTML prototype with 19 screened tiles and 7 ranked communities.
- Confirmed the working stack: Scopus AI, ASEAN/ITU evidence, Overpass Turbo, GeoLine, Colab/Python, GeoPandas, Rasterio, and Folium.
- Applied responsible checks: claim boundaries, sensitivity analysis, and data-gap awareness.
- Avoided overclaiming coverage or infrastructure deployment decisions.

Transition to Session 7 — FROM GEOAI INSIGHTS TO IMPACT

VALUE & IMPACT
Translate GeoAI outputs into decision value
Connect the toolkit, evidence, and prototype to outcomes, beneficiaries and potential impact.
Canvas 9

→

OPERATIONALISE
Canvas 9–10 + Integration
Move from prototype toward real-world implementation
Define validation, authoritative data, legal assessment and the implementation pathway.
Canvas 10

→

COMMUNICATE
Turn to ne credible journey into a defensible story
Convey evidence, problem, evidence, GeoAI approach, result, limitations and next action.
Integration

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THANK YOU

Bootcamp and Hackathon for ASEAN Geospatial Innovation 2026